

Security Patrol Monitoring System

Panduan Deployment Lengkap

Backend (Laravel 12 + MySQL 8) · Web Dashboard (Vue 3) · Docker (Compose) ·
Aplikasi Android (Kotlin)

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github.com/aykanesia/security-patrol

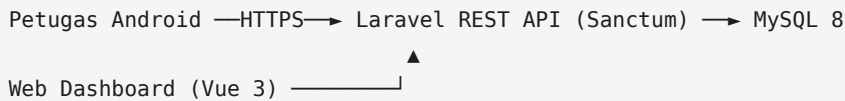
Dokumen ini merangkum seluruh panduan deployment: instalasi klasik (Nginx + PHP-FPM + MySQL), deployment otomatis dengan Docker Compose, serta build & distribusi aplikasi Android untuk petugas security.

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1. Ringkasan Sistem & Arsitektur

Security Patrol Monitoring System adalah aplikasi monitoring patroli petugas keamanan perumahan berbasis QR Code + GPS. Sistem terdiri dari REST API (Laravel 12 + Sanctum + MySQL 8), web dashboard (Vue 3 + Vite + Bootstrap 5 + Pinia + Leaflet), dan aplikasi Android native (Kotlin) untuk petugas security. Seluruh komponen berbagi satu API yang sama.



Fitur utama

- 3 peran: super_admin, supervisor, security (web untuk admin/supervisor, Android untuk security).
- Master data: area, checkpoint (QR token random), rute (SEQUENTIAL/FLEXIBLE), jadwal + penugasan, user, device.
- Alur patroli: login → jadwal hari ini → mulai patroli (GPS) → scan QR di tiap checkpoint → selesai.
- Validasi keamanan server-side: QR token unik, GPS Haversine, timestamp server, urutan (sequential), anti duplikat.
- Idempotensi UUID per check-in; dukungan offline sync (POST /sync) dari Android.
- Web dashboard: statistik, monitoring patroli aktif + peta Leaflet, histori, laporan + export, audit log, notifikasi.

Struktur repository

```

security-patrol/
├── backend/      Laravel 12 (REST API v1, Sanctum, MySQL) + 33 test
├── frontend/     Vue 3 + Vite + Bootstrap 5 + Pinia + Leaflet
├── android/      Aplikasi Android Kotlin (Retrofit, QR ZXing, GPS, offline sync)
├── docs/         API_SPEC.md · DEPLOYMENT.md · DOCKER.md · ANDROID_BUILD.md
├── docker-compose.yml  Docker: MySQL + PHP-FPM + Nginx (SPA) + scheduler
├── .env.example   Konfigurasi docker compose (salin ke .env)
└── scripts/      smoke-test.sh · e2e-patrol-test.sh · mysql*.sh
  
```

2. Persiapan & Prasyarat

Server produksi (backend + web)

Kebutuhan	Spesifikasi
RAM/CPU	Min. 2 GB RAM / 2 vCPU
PHP (manual)	8.3 + ekstensi: mbstring, xml, curl, mysql, zip, bcmath, intl, gd
Database	MySQL 8 (utf8mb4)
Web server (manual)	Nginx + PHP-FPM
Docker (opsional)	Docker Engine + Docker Compose plugin
Node.js	20+ (hanya untuk build frontend)
Composer	2.x (hanya untuk install backend manual)
Domain/HTTPS	Disarankan (certbot untuk SSL)

Akun demo (hasil seed — WAJIB ganti password setelah masuk!)

Username	Password	Role	Akses
supervisor	password	supervisor	Web dashboard (monitoring, laporan)
budi / andi / citra	password	security	Aplikasi Android

Catatan: aplikasi Android hanya menerima login role security. device_uuid dibuat otomatis per instalasi dan terikat ke akun.

3. Opsi Deployment (Manual vs Docker)

Ada dua cara menjalankan backend + web dashboard di server. Pilih sesuai kondisi; keduanya menghasilkan layanan yang sama untuk aplikasi Android (cukup arahkan API_BASE_URL ke domain/IP server).

Aspek	Manual (Nginx + PHP-FPM)	Docker Compose
Instalasi	Apt + composer + npm (banyak langkah)	docker compose up -d --build
Isolasi	Bergantung paket sistem	Penuh (container per service)
Update	composer/npm + config cache manual	git pull + docker compose up -d --build
Panduan	Bagian 4 dokumen ini	Bagian 5 dokumen ini
Android	Arahkan ke https://domain/api/v1	Sama — lewat reverse proxy/port host

4. Deployment Backend + Web — Manual (Nginx + PHP-FPM)

Panduan ini mengikuti docs/DEPLOYMENT.md. Target: Linux (Ubuntu/Debian 22.04+) + Nginx + PHP 8.3-FPM + MySQL 8 + Node.js. Satu domain melayani web dashboard (static Vue) dan API Laravel.

4.1 Instalasi dependensi server

```
# update sistem
sudo apt update && sudo apt upgrade -y

# install dependensi
sudo apt install -y nginx mysql-server php8.3-fpm php8.3-cli \
  php8.3-mbstring php8.3-xml php8.3-curl php8.3-mysql php8.3-zip \
  php8.3-bcmath php8.3-intl php8.3-gd unzip git curl

# composer
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer

# node (untuk build frontend)
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

4.2 Siapkan database

```
CREATE DATABASE security_patrol CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'patrol'@'localhost' IDENTIFIED BY 'GANTI_PASSWORD_KUAT';
GRANT ALL PRIVILEGES ON security_patrol.* TO 'patrol'@'localhost';
FLUSH PRIVILEGES;
```

4.3 Upload & setup backend

```
sudo mkdir -p /var/www/patroli && cd /var/www/patroli
# upload isi folder backend/ ke sini (scp / git clone), lalu:
cd backend
composer install --no-dev --optimize-autoloader
cp .env.example .env && nano .env
```

Isi .env yang wajib diubah:

```
APP_NAME="Security Patrol"
APP_ENV=production
APP_DEBUG=false
APP_URL=https://patroli.domain.com
APP_LOCALE=id
APP_TIMEZONE=Asia/Jakarta

DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=security_patrol
DB_USERNAME=patrol
DB_PASSWORD=GANTI_...KUAT

SESSION_DRIVER=database
QUEUE_CONNECTION=database
CACHE_STORE=database
```

```
# generate key + migrasi + seed
php artisan key:generate
php artisan migrate --seed

# optimasi produksi
php artisan config:cache && php artisan route:cache && php artisan view:cache
php artisan storage:link
```

4.4 Build frontend (Vue)

```
cd /var/www/patroli/frontend
npm ci
npm run build          # hasil → dist/
# bila env server memaksa NODE_ENV=production:
NODE_ENV=development npm run build
```

4.5 Konfigurasi Nginx

Buat /etc/nginx/sites-available/patroli (ringkas):

```

server {
    listen 80;
    server_name patroli.domain.com;
    return 301 https://$host$request_uri;
}
server {
    listen 443 ssl http2;
    server_name patroli.domain.com;
    ssl_certificate      /etc/letsencrypt/live/patroli.domain.com/fullchain.pem;
    ssl_certificate_key  /etc/letsencrypt/live/patroli.domain.com/privkey.pem;
    root /var/www/patroli/frontend/dist;
    index index.html;
    client_max_body_size 10M;

    # API Laravel
    location /api/ {
        root /var/www/patroli/backend/public;
        try_files $uri /index.php?$query_string;
        location ~ /\.php$ {
            include snippets/fastcgi-php.conf;
            fastcgi_pass unix:/var/run/php/php8.3-fpm.sock;
            fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        }
    }
    # health check
    location /up {
        root /var/www/patroli/backend/public;
        try_files /up /index.php?$query_string;
        location ~ /\.php$ {
            include snippets/fastcgi-php.conf;
            fastcgi_pass unix:/var/run/php/php8.3-fpm.sock;
            fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        }
    }
    # SPA Vue
    location / { try_files $uri $uri/ /index.html; }
    location ~* \.(js|css|woff2?|png|jpg|svg)$ {
        expires 30d;
        add_header Cache-Control "public, immutable";
    }
}

```

Aktifkan + SSL:

```

sudo ln -s /etc/nginx/sites-available/patroli /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d patroli.domain.com

```

4.6 Permission & scheduler

```

sudo chown -R www-data:www-data /var/www/patroli/backend/storage
sudo chown -R www-data:www-data /var/www/patroli/backend/bootstrap/cache
sudo crontab -u www-data -e

```

```

* * * * * cd /var/www/patroli/backend && php artisan schedule:run >> /dev/null 2>&1

```

4.7 Verifikasi

```
curl https://patroli.domain.com/up # → 200
curl https://patroli.domain.com/api/v1/me # → 401 JSON (normal)
curl -X POST https://patroli.domain.com/api/v1/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username":"admin","password":"password"}' # → token
```

5. Deployment Backend + Web — Docker Compose

File Docker berada satu repo dengan aplikasi (Opsi A), sehingga versi Docker selalu sinkron dengan kode. Empat service dijalankan sekaligus dengan satu perintah. Panduan lengkap: docs/DOCKER.md.

5.1 Struktur file Docker

```
security-patrol/
├── docker-compose.yml      orkestrasi 4 service
├── .env.example            template konfigurasi (→ .env)
├── backend/
│   ├── Dockerfile         PHP 8.3-FPM + composer + laravel
│   └── docker/entrypoint.sh APP_KEY persist, tunggu DB, migrasi+seed, cache
└── frontend/
    ├── Dockerfile         2-stage: node build → nginx SPA
    └── docker/nginx.conf   SPA Vue + proxy /api → backend:9000
```

5.2 Service (docker-compose.yml)

Service	Image/Build	Peran
backend	build ./backend	PHP-FPM Laravel; entrypoint migrasi + seed
scheduler	build ./backend	php artisan schedule:work (notifikasi patroli, tiap 5 mnt)
frontend	build ./frontend	Nginx: serve dist/ Vue + proxy /api/* → backend

5.3 Menjalankan

```
# 1) siapkan konfigurasi
cp .env.example .env
nano .env          # isi DB_PASSWORD & DB_ROOT_PASSWORD dengan password kuat

# 2) bangun & jalankan (pertama kali ~3-5 menit)
docker compose up -d --build

# 3) pantau log sampai seed selesai
docker compose logs -f backend

# 4) verifikasi
curl http://localhost:8080/up          # → 200
curl http://localhost:8080/api/v1/me  # → 401 JSON (normal)
```

Buka <http://localhost:8080> → halaman login web dashboard. Seed hanya berjalan sekali (marker storage/.seeded di volume); reset total dengan `docker compose down -v` lalu up lagi.

5.4 Contoh .env untuk Docker

```
APP_PORT=8080
APP_URL=http://localhost:8080
DB_DATABASE=security_patrol
DB_USERNAME=patrol
DB_PASSWORD=GANTI_PASSWORD_DB_KUAT
DB_ROOT_PASSWORD=GANTI_PASSWORD_ROOT_KUAT
APP_KEY=          # kosong = auto-generate & persist di volume
```

5.5 Perintah sehari-hari & update

```

docker compose ps           # status service
docker compose logs -f backend # log Laravel
docker compose logs -f scheduler # log scheduler
docker compose restart backend # restart satu service
docker compose down         # stop (data volume TETAP)
docker compose down -v      # stop + HAPUS volume (db & seed hilang)

# update ke versi baru:
git pull && docker compose up -d --build

```

5.6 Kustomisasi penting

Hal	Lokasi	Catatan
Password DB	.env → DB_PASSWORD / DB_ROOT_PASSWORD	wajib diganti
URL publik	.env → APP_URL	untuk generate link
APP_KEY	.env → APP_KEY	kosong = otomatis (persist di volume)
Zona waktu	compose → APP_TIMEZONE	default Asia/Jakarta
Versi PHP	backend/Dockerfile	php:8.3-fpm
Versi Node	frontend/Dockerfile	node:22-alpine

Container tidak memakai php artisan serve — Nginx terhubung langsung ke PHP-FPM (backend:9000) seperti arsitektur produksi. Untuk produksi dengan domain, pasang reverse-proxy HTTPS di host (mis. Caddy/Nginx) dan set APP_URL=https://domain.

6. Aplikasi Android — Build & Distribusi

Folder android/ berisi project Android Studio (Kotlin) untuk petugas security. Lingkungan build asisten tidak memiliki Java/Android SDK, sehingga source dikirim lengkap & teraudit — buka di Android Studio di

komputer Anda, build, lalu instal ke HP. Panduan lengkap: docs/ANDROID_BUILD.md.

6.1 Fitur aplikasi

Fitur	Lokasi (Kotlin)
Jadwal hari ini + mulai patroli (GPS)	ui/home/ScheduleFragment.kt
Riwayat + detail patroli	ui/home/HistoryFragment.kt, ui/patrol/PatrolDetailActivity.kt
Patroli berjalan: progress, scan QR, selesai/batal	ui/patrol/PatrolActivity.kt
QR scanner (ZXing embedded)	PatrolActivity.launchScanner()
GPS (FusedLocationProvider)	util/LocationHelper.kt
DTO + Retrofit + error envelope	data/remote/*
Antrian offline + sync berkala 15 mnt	data/local/OfflineQueue.kt, data/sync/SyncWorker.kt
Token & sesi aman	data/local/SessionManager.kt

6.2 Prasyarat

- Android Studio terbaru (Ladybug+) — unduh dari developer.android.com/studio
- Saat sync pertama, Android Studio mengunduh otomatis: JDK 17, Android SDK (compileSdk 35), Gradle

8.9 (wrapper disertakan).

- Koneksi internet stabil (sekali unduh ~1-2 GB).

6.3 Buka project & set URL API

```
# Buka: File → Open → pilih folder android/ → tunggu Gradle sync.
# Struktur penting:
android/app/build.gradle.kts ← SDK, dependency, API_BASE_URL
android/app/src/main/        ← manifest, kotlin, res
```

Base URL diatur di android/app/build.gradle.kts:

```
defaultConfig {
    // Release: URL produksi
    buildConfigField("String", "API_BASE_URL", "\"https://patroli.domain.com/api/v1\"")
}
buildTypes {
    debug {
        // Emulator: 10.0.2.2 = localhost host
        buildConfigField("String", "API_BASE_URL", "\"http://10.0.2.2:8080/api/v1\"")
    }
}
```

Skenario	API_BASE_URL
Backend di Docker (host sama)	http://10.0.2.2:8080/api/v1 (APP_PORT)
HP asli satu jaringan	http://IP-SERVER:8080/api/v1
Produksi (domain)	https://patroli.domain.com/api/v1

Catatan: usesCleartextTraffic=true diizinkan untuk HTTP saat dev; untuk produksi gunakan HTTPS.

6.4 Build & install

```
# Dari Android Studio:
# 1. Colok HP (Developer options → USB debugging) atau nyalakan emulator
# 2. Klik Run ► (Shift+F10) → pilih device
#
# APK manual via baris perintah:
cd android
./gradlew assembleDebug      # → app/build/outputs/apk/debug/app-debug.apk
./gradlew assembleRelease    # butuh signing config (§6.5)
```

6.5 Persiapan release (Play Store / distribusi APK)

- Ganti base URL ke produksi (HTTPS).
- Buat keystore & signingConfig di app/build.gradle.kts (storeFile release.keystore, alias, password).
- Jalankan ./gradlew assembleRelease; hasil APK di app/build/outputs/apk/release/.
- Disarankan aktifkan isMinifyEnabled = true + proguard untuk produksi.

6.6 Alur scan offline & catatan QR

Saat tidak ada internet, hasil scan disimpan lokal (OfflineQueue) lalu dikirim otomatis ke POST /sync oleh

WorkManager begitu koneksi pulih (atau tiap 15 menit). Item sukses/duplikat dihapus dari antrian; item gagal bisnis ditampilkan sebagai error. Backend menerima scan_code berupa kode publik (CP001) atau QR token penuh (PATROL:CP001:8f92a7c1...). QR yang ditempel di checkpoint sebaiknya token penuh (ambil via GET /admin/checkpoints/{id}/qr).

7. Lampiran A — Daftar Endpoint API Ringkas

Base URL: <https://patroli.domain.com/api/v1> (produksi) atau <http://127.0.0.1:8000/api/v1> (dev). Seluruh request memakai HTTPS + format JSON. Auth: Laravel Sanctum (Bearer token).

7.1 Autentikasi

Method & Path	Akses	Fungsi
POST /auth/logout	auth	Hapus token aktif
GET /me	auth	Profil user

7.2 Patroli (role security)

Method & Path	Fungsi
POST /patrol/start	Mulai patroli (schedule_id + lat/lon)
GET /patrol/current	Session RUNNING + daftar checkpoint + progress
POST /patrol/checkpoint/scan	Scan QR/checkpoint (uuid, lat/lon)
POST /patrol/complete	Selesaikan patroli (wajib checkpoint lengkap)
POST /patrol/cancel	Batal session RUNNING
GET /patrol/history	Histori session (pagination + filter)
GET /patrol/detail/{sessionCode}	Detail session + daftar check-in
POST /sync	Offline sync batch (idempoten per UUID)

7.3 Dashboard, laporan, notifikasi (super_admin/supervisor)

Method & Path	Fungsi
GET /dashboard/active-patrols	Patroli RUNNING + progress
GET /dashboard/officer-positions	Posisi petugas utk peta Leaflet
GET /sessions · GET /sessions/{id}	Histori & detail semua session
POST /sessions/{id}/incomplete	Tandai INCOMPLETE + alasan (supervisor)
GET /reports/daily · /monthly · /attendance	Laporan harian/bulanan/kehadiran
GET /reports/export/daily · /range	Export CSV/Excel
GET /notifications · POST .../read · read-all	Notifikasi user

7.4 Admin CRUD (role super_admin, prefix /admin)

Method & Path	Fungsi
areas · checkpoints	CRUD area & checkpoint (qr_token otomatis)
checkpoints/{id}/qr	Ambil QR token checkpoint
routes	CRUD rute (SEQUENTIAL/FLEXIBLE) + checkpoint_ids
schedules	CRUD jadwal + user_ids
devices · {id}/block · {id}/unblock	Manajemen device petugas
audit-logs	Riwayat audit perubahan

8. Lampiran B — Kode Error API

Envelope sukses: {success:true, message, data, meta}. Envelope error: {success:false, message, error_code, data}. Validasi Laravel (422) mengembalikan {message, errors}.

error_code	HTTP	Arti
FORBIDDEN	403	Role tidak punya akses
INVALID_LOCATION	422	GPS di luar radius checkpoint
INVALID_CHECKPOINT	422	QR/kode tidak dikenali
INVALID_SESSION	403/422	Session bukan milik user / tidak RUNNING
INVALID_SEQUENCE	422	Urutan checkpoint salah (rute SEQUENTIAL)
DUPLICATE_CHECKIN	422	Checkpoint sudah discan valid
ALREADY_PROCESSED	200	UUID sama sudah diproses (idempoten)
CHECKPOINT_INCOMPLETE	422	Complete ditolak — checkpoint belum lengkap
SESSION_ALREADY_RUNNING	422	Masih ada patroli berjalan
SCHEDULE_TOO_EARLY / TOO_LATE	422	Di luar window jadwal (grace)
DEVICE_REQUIRED / BLOCKED / OWNED	422/403	Validasi device

9. Lampiran C — Checklist Go-Live & Keamanan

9.1 Checklist go-live (manual)

- ☐ APP_DEBUG=false dan APP_ENV=production
- ☐ Password default (admin/supervisor/budi) diganti
- ☐ HTTPS aktif (certbot / reverse proxy)
- ☐ DB user memakai password kuat, bukan default
- ☐ Firewall: buka port 80/443 saja
- ☐ Backup rutin: mysqldump security_patrol + folder backend/storage
- ☐ Cron scheduler berjalan (notifikasi patroli)
- ☐ Uji: GET /up → 200; POST /auth/login → token; web dashboard terbuka

9.2 Checklist go-live (Docker)

- ☐ Isi .env: DB_PASSWORD & DB_ROOT_PASSWORD kuat, APP_URL benar
- ☐ docker compose up -d --build sukses; seed selesai (log backend)
- ☐ Uji curl http://host:8080/up dan /api/v1/me (401 JSON)
- ☐ Pasang reverse-proxy HTTPS untuk produksi; set APP_URL=https://...
- ☐ Backup volume: mysql_data + backend_storage (atau mysqldump di dalam container)
- ☐ Pantau docker compose logs (backend & scheduler) secara berkala

9.3 Checklist Android

- ☐ API_BASE_URL sesuai environment (emulator/HP/server)
- ☐ Debug build berhasil di Android Studio; login pakai akun role security
- ☐ GPS HP akurat (radius checkpoint ≥ 25-30 m agar scan tidak ditolak)
- ☐ Tes offline: matikan data → scan → nyalakan → tunggu sync otomatis
- ☐ Release: signing config + HTTPS + ganti usesCleartextTraffic bila perlu

9.4 Catatan keamanan

- QR token random & unik — tidak berisi data sensitif, sulit ditebak.
- GPS diverifikasi backend (Haversine) — manipulasi client tidak berguna.
- Timestamp resmi = server clock (scanned_at); device_timestamp hanya referensi.
- Token Sanctum: selalu HTTPS; jangan hardcode di APK.
- Segera revoke token akses GitHub yang pernah bocor & ganti password akun.
- Scan gagal validasi tetap tercatat (INVALID_*, sync_status=FAILED) untuk audit.

9.5 Troubleshooting ringkas

Gejala	Solusi
Backend restart (Docker)	docker compose logs backend — umumnya DB belum siap; entripoint menunggu
419 Page Expired	Pastikan SESSION_DRIVER=database & sudah migrate
403 storage	chown -R www-data storage
Scan selalu di luar radius	GPS HP kurang akurat / radius checkpoint kecil
Login Android langsung kembali	Pastikan akun role security
Emulator tak dapat API	Jalankan backend dgn --host=0.0.0.0 (atau Docker) & pakai 10.0.2.2

— Akhir dokumen. Seluruh panduan rinci tersedia di docs/ (API_SPEC.md, DEPLOYMENT.md, DOCKER.md, ANDROID_BUILD.md).